Table 6: Examples of heuristic neurons for the multiplication operator

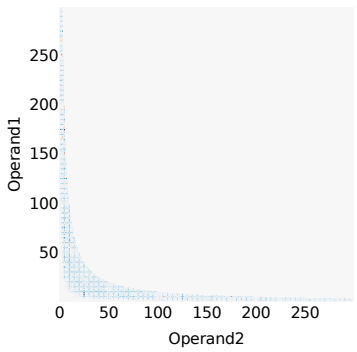
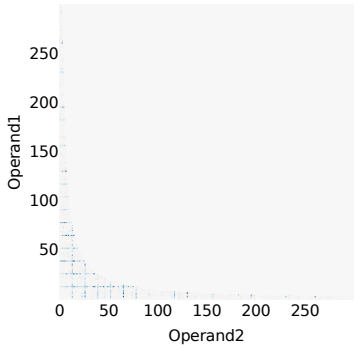
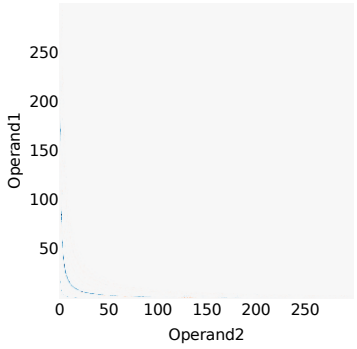
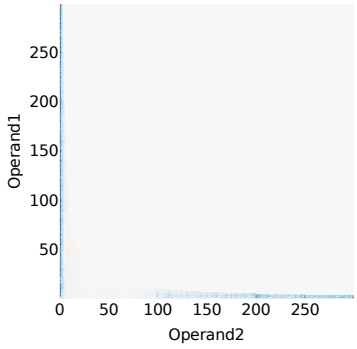
Neuron number $\mathbf{h}_{post}^{l,n}$	Activation pattern	Top-10 numerical logits in value vector $\mathbf{v}_{out}^{l,n}$	Classified Heuristics
$\mathbf{h}_{post}^{31,8084}$		'320', '360', '770', '630', '820', '510', '420', '240', '370', '380'	$result \equiv 0 \pmod 5$, pattern_{result} ..5, pattern_{op2} 010
$\mathbf{h}_{post}^{30,9988}$		'585', '936', '546', '637', '975', '455', '910', '819', '715', '364'	$op1 \equiv 0 \pmod{13}$, $op2 \equiv 0 \pmod{13}$, $result \equiv 0 \pmod{13}$
$\mathbf{h}_{post}^{21,12521}$		'170', '162', '171', '169', '17', '156', '160', '16', '785', '168'	$\text{range}_{result} \in [150, 180]$, $result \equiv 0 \pmod 2$
$\mathbf{h}_{post}^{16,2956}$		'844', '848', '873', '985', '548', '858', '657', '995', '788', '716'	pattern_{op2} 0.1

Table 7: Examples of heuristic neurons for the division operator

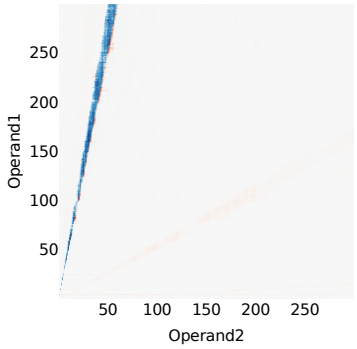
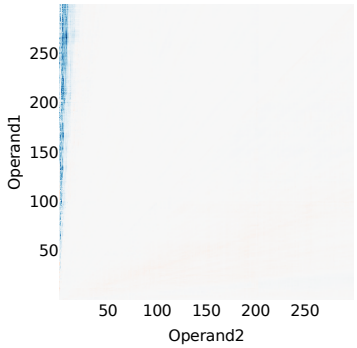
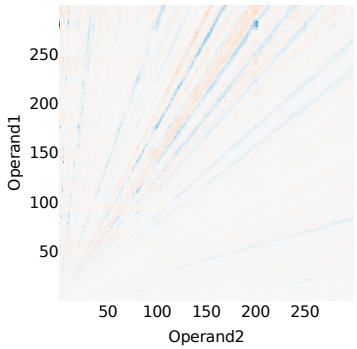
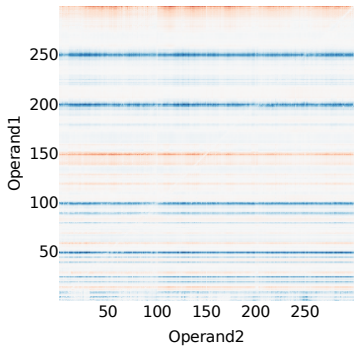
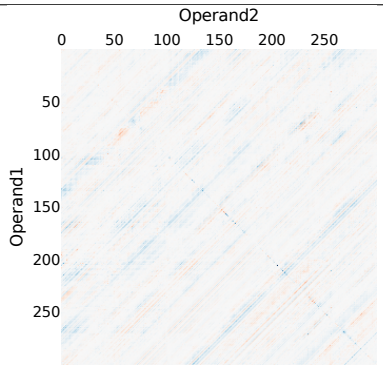
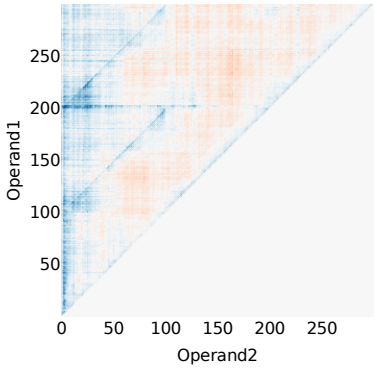
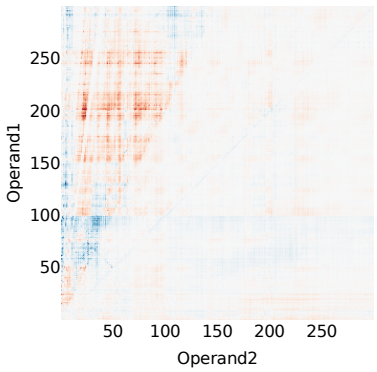
Neuron number $\mathbf{h}_{post}^{l,n}$	Activation pattern	Top-10 numerical logits in value vector $\mathbf{v}_{out}^{l,n}$	Classified Heuristics
$\mathbf{h}_{post}^{30,3168}$		'5', '005', '05', '948', '891', '349', '324', '544', '905', '801'	pattern_{result} 005, $\text{range}_{result} \in [4-6]$
$\mathbf{h}_{post}^{25,1402}$		'23', '33', '38', '08', '22', '00', '26', '30', '07', '32'	$\text{range}_{op2} \in [0, 10]$, $\text{range}_{result} \in [0, 100]$, pattern_{result} ..3
$\mathbf{h}_{post}^{24,9306}$		'8', '446', '770', '7', '740', '674', '449', '741', '08', '433'	multi-value$_{result} \in \{1, 2, 7, 8\}$
$\mathbf{h}_{post}^{17,10704}$		'568', '316', '773', '405', '621', '822', '582', '549', '587', '598',	pattern_{op1} ..0, pattern_{op1} .50

Table 8: Examples of neurons that failed to classify as one of the defined heuristic types

Neuron number $\mathbf{h}_{post}^{l,n}$	Activation pattern	Top-10 numerical logits in value vector $\mathbf{v}_{out}^{l,n}$
$\mathbf{h}_{post}^{22,14024}$, Addition		’844’, ’839’, ’568’, ’216’, ’636’, ’877’, ’786’, ’832’, ’084’, ’536’
$\mathbf{h}_{post}^{19,12469}$, Subtraction		’344’, ’3’, ’444’, ’509’, ’4’, ’456’, ’445’, ’2’, ’333’, ’5’
$\mathbf{h}_{post}^{27,4900}$, Multiplication		’270’, ’409’, ’257’, ’454’, ’451’, ’570’, ’290’, ’287’, ’470’, ’439’
$\mathbf{h}_{post}^{20,11020}$, Division		’823’, ’983’, ’801’, ’731’, ’501’, ’751’, ’070’, ’663’, ’985’, ’713’